



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Quadratic Graphs

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What is a quadratic graph?

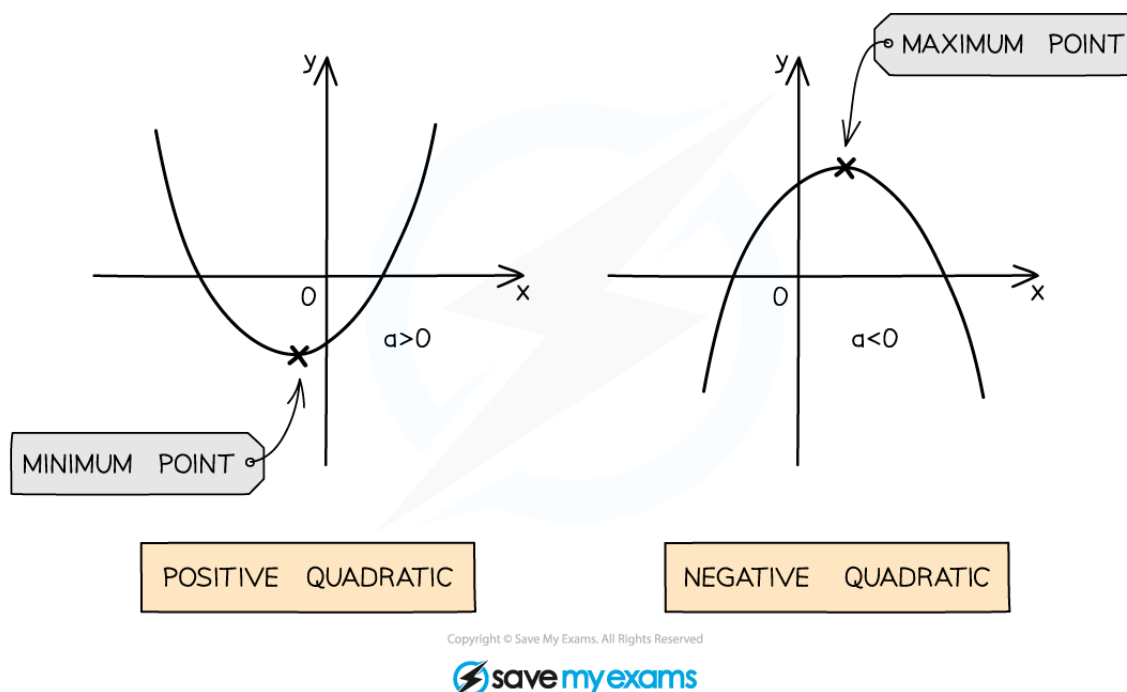
- A **quadratic graph** has the form $y = ax^2 + bx + c$
 - where a is not zero

What does a quadratic graph look like?

- A quadratic graph is a smooth curve with a **vertical line of symmetry**
 - A **positive** number in front of x^2 gives a **u-shaped curve**
 - A **negative** number in front of x^2 gives an **n-shaped curve**
- The shape made by a quadratic graph is known as a **parabola**
- A quadratic graph will **always** cross the **y-axis**
- A quadratic graph crosses the **X-axis twice, once, or not at all**
 - The points where the graph crosses the **X-axis** are called the **roots**
- If the graph is a **u-shape**, it has a **minimum point**
- If the graph is an **n-shape**, it has a **maximum point**
- Minimum and maximum points are both examples of **turning points**
 - A turning point can also be called a **vertex**



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How do I sketch a quadratic graph?

- It is important to know how to **sketch** a quadratic curve
 - A simple drawing showing the **key features** is often sufficient
 - (For a more accurate graph, create a table of values and plot the points)
- To **sketch a quadratic graph**:
 - First sketch the **X** and **y**-axes
 - Identify the **y**-intercept and mark it on the **y**-axis
 - The **y**-intercept of $y = ax^2 + bx + c$ will be $(0, c)$
 - It can also be found by substituting in $x = 0$
 - Find all root(s) (0, 1 or 2) of the equation and mark them on the **X**-axis
 - The roots will be the solutions to $y = 0$; $ax^2 + bx + c = 0$
 - You can find the solutions by factorising, completing the square or using the quadratic formula



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- Identify if the number a in $ax^2 + bx + c$ is positive or negative
 - A positive value will result in a u-shape
 - A negative value will result in an n-shape
- Sketch a smooth curve through the X and Y -intercepts
 - Mark on any axes intercepts
 - Mark on the coordinates of the maximum/minimum point if you know it

How do I find the coordinates of the turning point by completing the square?

- The coordinates of the **turning point** (vertex) of a quadratic graph can be found by **completing the square**
- For a quadratic graph written in the form $y = a(x - p)^2 + q$
 - the minimum or maximum point has **coordinates** (p, q)
- Beware: there is a **sign change** for the X -coordinate
 - A curve with equation $y = (x - 3)^2 + 2$, has a minimum point at $(3, 2)$
 - A curve with equation $y = (x + 3)^2 + 2$, has a minimum point at $(-3, 2)$
- The value of a does **not affect the coordinates** of the turning point but it **will change the shape** of the graph
 - If it is positive, the graph will be a u-shape
 - The curve $y = 5(x - 3)^2 + 2$ has a **minimum** point at $(3, 2)$
 - Of it is negative, the graph will be an n-shape
 - The curve $y = -8(x - 3)^2 + 2$ has a **maximum** point at $(3, 2)$

How do I find the coordinates of the turning point using differentiation?

- The coordinates of the **turning point** (maximum/minimum) of a quadratic can be found through **differentiation**
- To find the coordinates of the turning point
 - Differentiate the quadratic equation $y = ax^2 + bx + c$



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- This will give you $\frac{dy}{dx}$
- Set $\frac{dy}{dx} = 0$ and solve for x
 - The solution will be the x -coordinate of the turning point
- Substitute the x value into $y = ax^2 + bx + c$
 - This will give you the y -coordinate of the turning point



Worked Example

(a) Sketch the graph of $y = x^2 - 5x + 6$ showing the x and y intercepts clearly.

The $+c$ at the end is the y -intercept

y -intercept: $(0, 6)$

Factorise the quadratic expression

$$y = (x - 2)(x - 3)$$

Solve $y = 0$

$$(x - 2)(x - 3) = 0, \text{ so } x = 2 \text{ or } x = 3$$

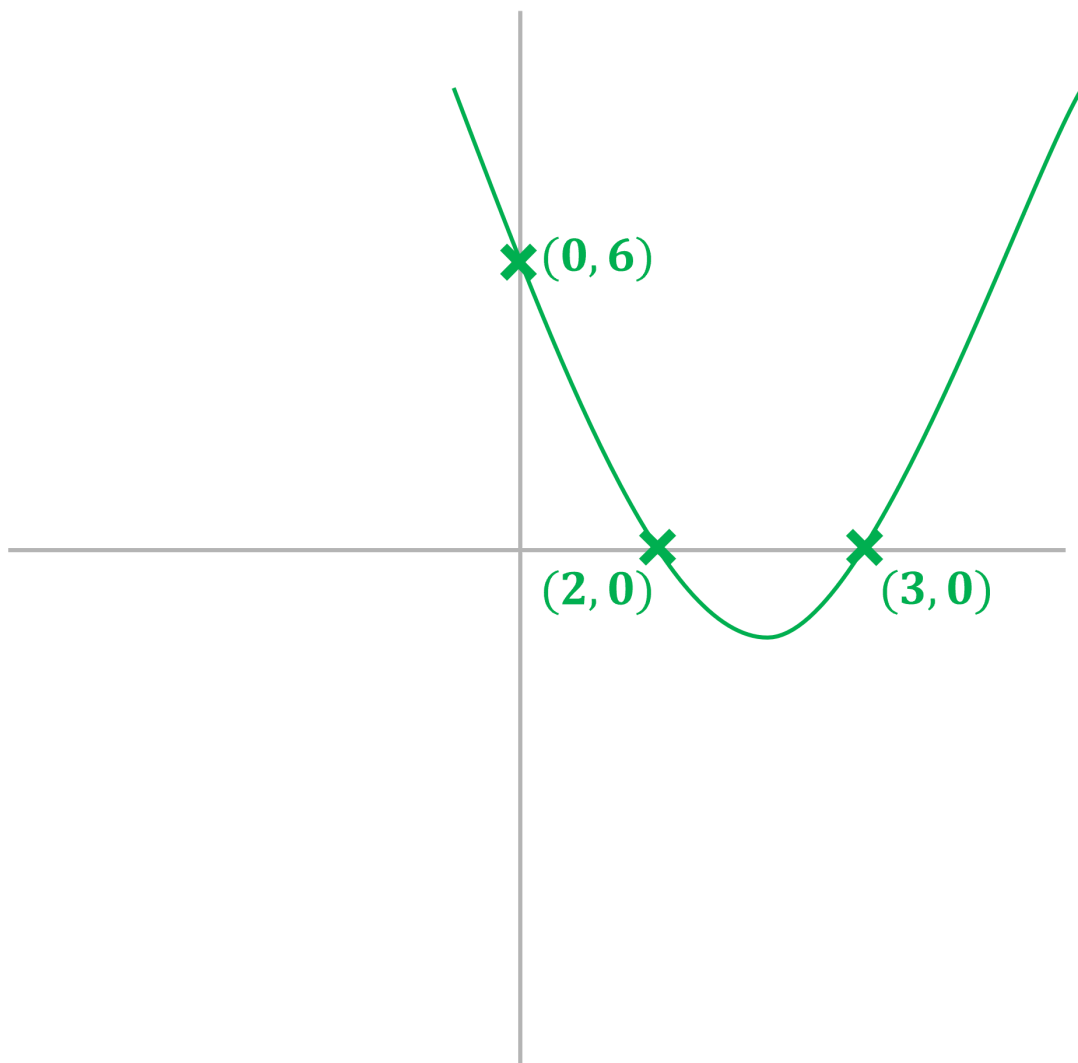
So the x -intercepts are given by the coordinates

$(2, 0)$ and $(3, 0)$

It is a positive quadratic graph, so will be a u-shape



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(b) Sketch the graph of $y = x^2 - 6x + 13$ showing the y -intercept and the coordinates of the turning point.

It is a positive quadratic, so will be a u-shape
The turning point will therefore be a minimum

The $+c$ at the end is the y -intercept

y -intercept: $(0, 13)$

Find the minimum point by completing the square (or through differentiation)



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For example, complete the square by writing the equation in the form $a(x - p)^2 + q$ (you may need to look this method up)

$$\begin{aligned}x^2 - 6x + 13 &= (x - 3)^2 - 9 + 13 \\ &= (x - 3)^2 + 4\end{aligned}$$

The turning point of $y = a(x - p)^2 + q$ has coordinates (p, q)

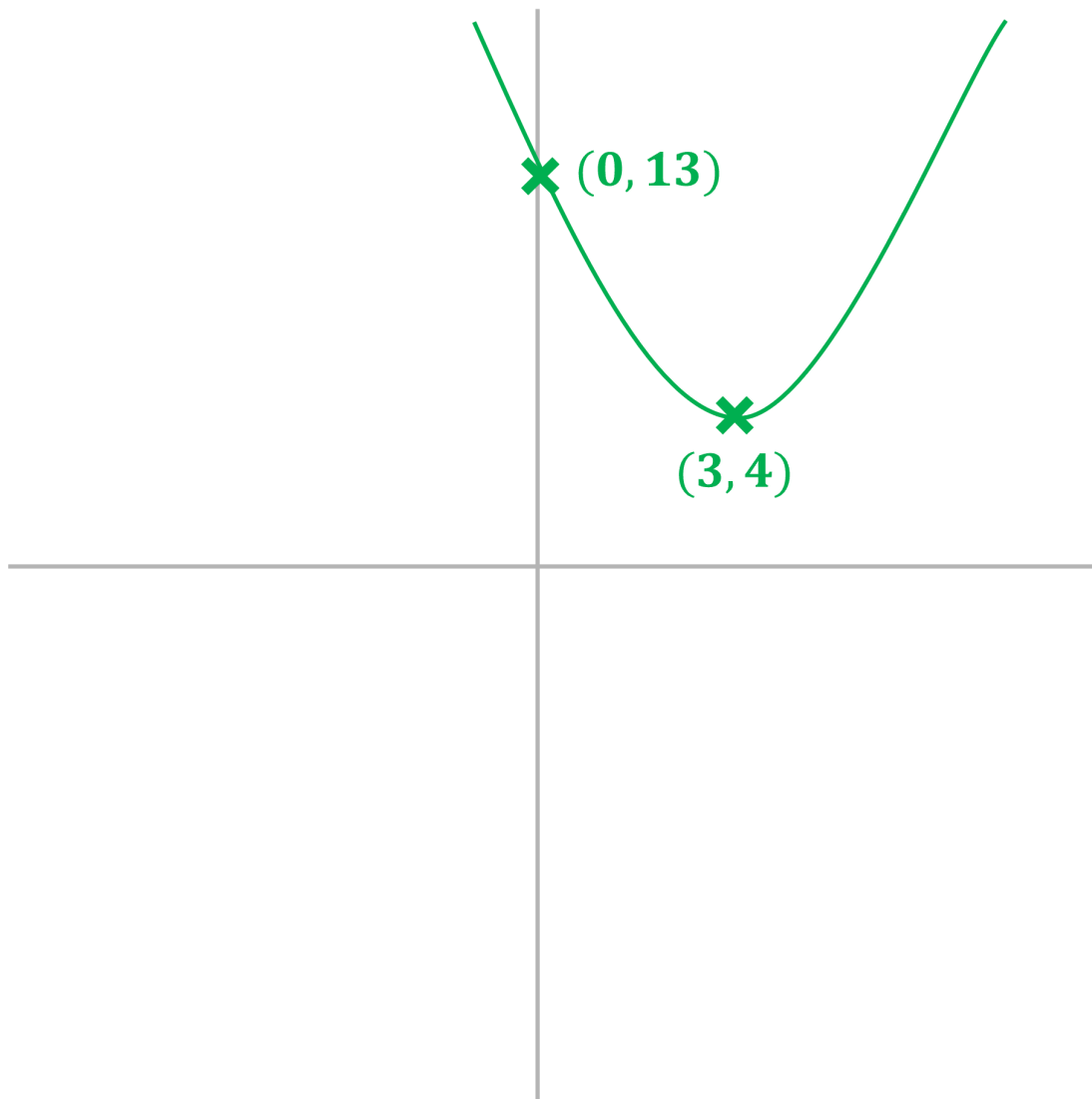
The minimum point is therefore

$$(3, 4)$$

As the **minimum** point is above the X -axis, and the curve is a u-shape, this means the graph will not cross the X -axis (it has no roots)



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(c) Sketch the graph of $y = -x^2 - 4x - 4$ showing the root(s), y -intercept, and the coordinates of the turning point.

It is a negative quadratic, so will be an n-shape
The turning point will therefore be a maximum

The $+c$ at the end is the y -intercept

y -intercept: $(0, -4)$



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Find the minimum point by completing the square or through differentiation

For example, differentiate the equation

$$\frac{dy}{dx} = -2x - 4$$

Set the derivative equal to zero and solve for x

$$0 = -2x - 4$$

$$2x = -4$$

$$x = -2$$

Substitute this value of x back into the original equation for y

$$y = -(-2)^2 - 4(-2) - 4$$

$$y = -4 + 8 - 4$$

$$y = 0$$

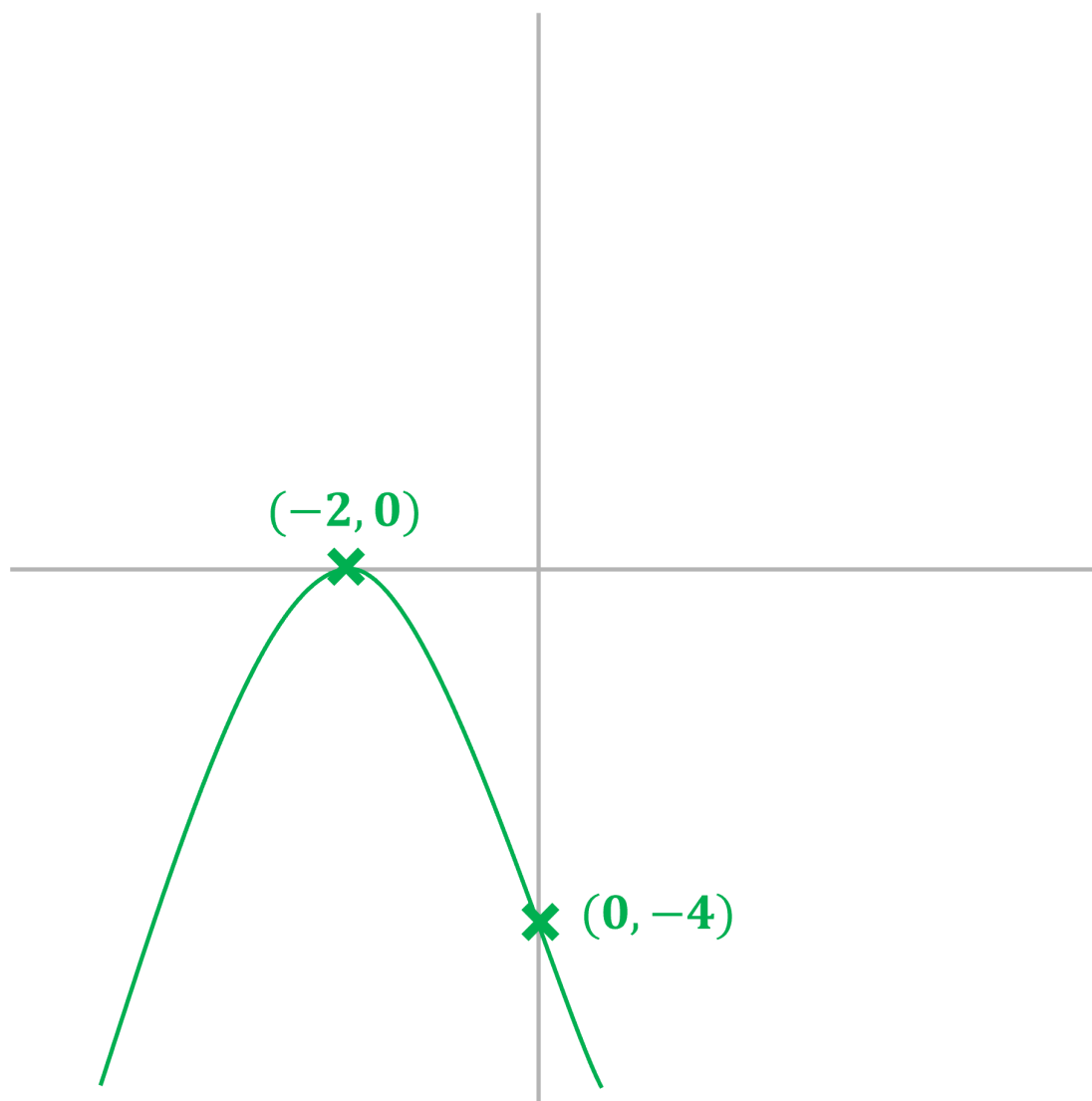
This shows that the maximum point has coordinates

$$(-2, 0)$$

As the maximum is on the x -axis, there is **only one root**



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How do I find the equation of a quadratic from its graph?

- If the **vertex** and **one other point** are known
 - Use the form $y = a(x - p)^2 + q$ to fill in p and q
 - The vertex is at (p, q)
 - Then substitute in the other known point (x, y) to find a
- If the **roots** (X -intercepts) and **one other point** are known



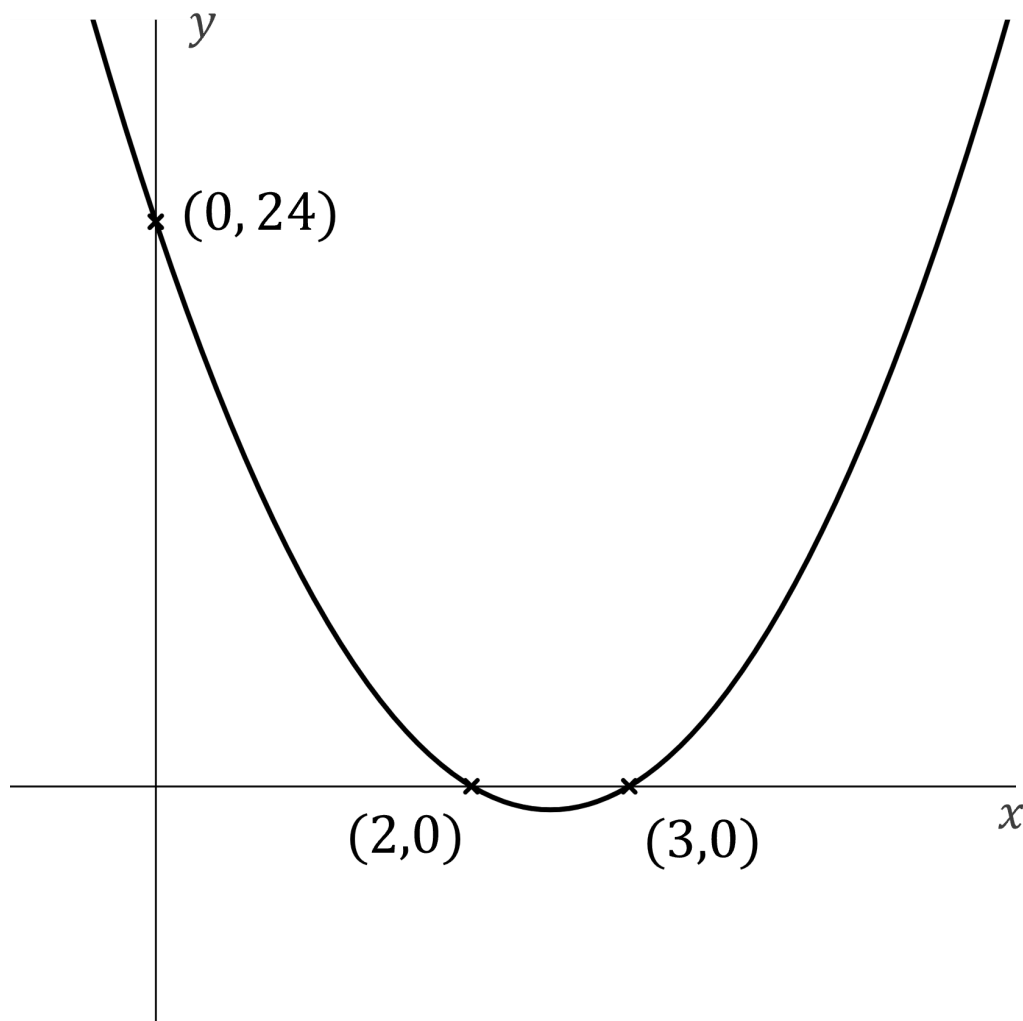
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- Use the form $y = a(x - x_1)(x - x_2)$ to fill in x_1 and x_2
 - The roots are at $(x_1, 0)$ and $(x_2, 0)$
 - Then substitute in the other known point (x, y) to find a
- If $a = 1$ then you only need either the **vertex** or the **roots**



Worked Example

(a) Find the equation of the graph below.





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The graph shows the roots and a point on the curve (in this case the y -intercept)

Use the form $y = a(x - x_1)(x - x_2)$ to fill in x_1 and x_2 by inspection

The roots are at $(x_1, 0)$ and $(x_2, 0)$

$$y = a(x - 2)(x - 3)$$

Substitute in the other known point $(0, 24)$ to find a

$$24 = a(0 - 2)(0 - 3)$$

$$24 = a(-2)(-3)$$

$$24 = 6a$$

$$4 = a$$

Write the full equation

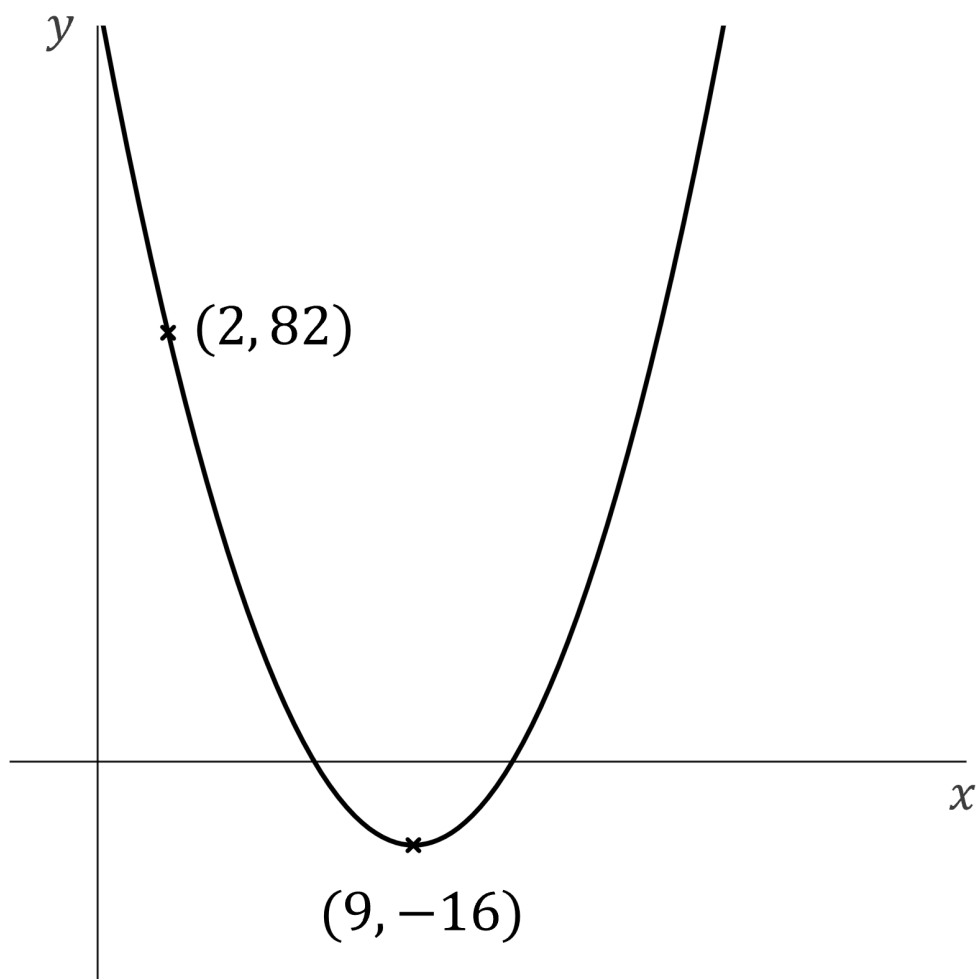
$$y = 4(x - 2)(x - 3)$$

You could also write this in expanded form: $y = 4x^2 - 20x + 24$

(b) Find the equation of the graph below.



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The graph shows the vertex and a point on the curve

Use the form $y = a(x - p)^2 + q$ to fill in p and q by inspection

The vertex is at (p, q)

$$y = a(x - 9)^2 - 16$$

Substitute in the other known point $(2, 82)$ to find a



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$$82 = a(2 - 9)^2 - 16$$

$$82 = a(-7)^2 - 16$$

$$82 = 49a - 16$$

$$98 = 49a$$

$$2 = a$$

Write the full equation

$$y = 2(x - 9)^2 - 16$$

You could also write this in expanded form: $y = 2x^2 - 36x + 146$